LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Daily Practice: Ch 10 The Human Eye and the Colourful World

Chapter / Topic: The Human Eye and the Colourful World · Week 5 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. One day at a time, about 25 minutes each: Day 1 on the day you study the chapter, then Day 2 and Day 3. Write answers in your notebook and check them with the key the same day. Copy every mistake into your error log.
2. Defects of vision: for myopia the lens makes a distant object's image at the far point; for hypermetropia it makes the image of an object at 25 cm at the near point. Then use the lens formula.

DAY 1: THE EYE AND ITS DEFECTS

Q1. Most of the refraction of light entering the eye takes place at the:

(a) eye lens
(b) outer surface of the cornea
(c) iris
(d) retina

Q2. The eye lens is:

(a) a concave lens
(b) a glass prism
(c) a flat, transparent sheet
(d) a convex lens made of a fibrous, jelly-like material

Q3. When we look at a distant object, the ciliary muscles are _____ and the eye lens becomes _____.

(a) relaxed, thin
(b) contracted, thick
(c) relaxed, thick
(d) contracted, thin

Q4. A person with hypermetropia needs spectacles with:

(a) concave lenses (negative power)
(b) lenses of zero power
(c) convex lenses (positive power)
(d) cylindrical lenses

Q5. What happens to the pupil when you walk from bright sunlight into a dark room? Why?

Q6. Why can we not see an object clearly when it is held closer than 25 cm to the eye?

Q7. A myopic person can see clearly only up to 80 cm. Find the nature and power of the lens needed to correct the defect.

Q8. Compare myopia and hypermetropia: what the person can see clearly, where the image of the problem object forms, and the lens used to correct each.

DAY 2: THE PRISM AND ATMOSPHERIC REFRACTION

Q9. The angle between the two refracting surfaces of a prism is called the:

(a) angle of the prism
(b) angle of incidence
(c) angle of deviation
(d) angle of refraction

Q10. A beam of white light passes through a glass prism. Which colour is deviated the least?

(a) violet
(b) red
(c) blue
(d) green

Q11. Newton showed that the colours of white light can be recombined into white light by using:

(a) a single prism
(b) a convex lens
(c) a concave mirror
(d) a second identical prism placed upside down

Q12. What is the angle of deviation in a prism? Describe the path of a ray through a triangular glass prism.

Q13. A rectangular glass slab does not split white light into colours, but a prism does. Why?

Q14. Objects seen through the hot air rising above a fire or a radiator seem to waver. Explain why.

DAY 3: SCATTERING, MIXED PRACTICE AND CHALLENGE

Q15. **Assertion (A):** A person with presbyopia may need bifocal lenses.
Reason (R): Presbyopia happens because the cornea becomes cloudy with age.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q16. The far point of a normal human eye is at:

(a) infinity (b) 25 cm
(c) 1 m (d) 100 m

Q17. The colour of scattered light depends mainly on:

(a) the temperature of the air
(b) the size of the scattering particles
(c) the speed of the wind
(d) the height of the observer

Q18. On a clear afternoon, Aman noticed that the sky was deep blue but the clouds were white.

(i) Which colour of sunlight is scattered most by the tiny molecules of air? Why?

(ii) Why do the clouds look white?

(iii) What is the Tyndall effect? Give one everyday example.

Q19. (a) What is hypermetropia? Give its two causes.
(b) The near point of a hypermetropic eye is 40 cm. Find the power of the lens needed to read at 25 cm.
(c) Why must this lens be convex?

Q20. The image formed on the retina is real and inverted. Why, then, do we see objects upright?

~ Week 5 done after this one. Physics is all about practice. Keep at it! ~